Seat No.

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 1 **Year:** 2024

Subject: 031001123 Chemistry III **Section:** 15-18

Date: 7 August 2024 **Time:** 08.00- 10.00

Name:________________________ID:____________________Class:_________

Directions: The test is designed to measure your comprehension. The test is divided into 3 sections. There will be 7 pages (including this page), 8 questions and they are worth 90 marks (40 points).

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators cannot be used in this test.

This exam paper is part of a set used to assess student abilities as follows:		Question (Q)
CLO 1	Explanation and comparison between covalent bond, ionic bond, and metallic bond.	Q1, Q2, Q3, Q5, Q7
CLO 2	Description of theoretical term and calculation of chemical bond energies.	Q6, Q8
CLO 3	Application of fundamental chemistry in other subjects.	Q4
CLO 4	Realization of discipline, punctuality, and public and self-responsibility.	

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Part1 : Total score 30 marks

1. Draw the Lewis structures (including the count of valence electrons and formal charges) of the following molecules: **(20 marks)**

Molecule/Ion	Total valence electron	Formal charges of central atom	Lewis structure
Methane (CH_4)			
Phosphine (PH_3)			
Carbon dioxide (CO_2)			
Sulfur dioxide (SO_2)			
Ammonium ion (NH_4^+)			
Phosphate ion (PO_4^{3-})			

2. Draw the Lewis structure of sulfite ion (SO_3^{2-}) and its resonance structures. **(10 marks)**

Part 2 : Total score 30 marks

3. Match the molecules in column A with the corresponding geometry shape in column B. Write the letter of your answer for each question. **(15 marks)**

Column A	Column B
.... 3.1) $SbBrF_4$	A. Trigonal planar
.... 3.2) NH_4^+	B. Trigonal bipyramidal
.... 3.3) $SeBr_2$	C. Octahedral
.... 3.4) SeI_6	D. Trigonal pyramidal
.... 3.5) SO_2	E. Square planar
.... 3.6) AsO_4^{3-}	F. Bent (angle < 109.5°)
.... 3.7) $SiOP^-$	G. Bent (angle between 109.5° – 120°)
.... 3.8) BF_3	H. Seesaw
.... 3.9) IF_3	I. T-Shaped
.... 3.10) $XeOF_3^+$	J. Square pyramidal
.... 3.11) XeF_2	K. Tetrahedral
.... 3.12) SF_4	L. Linear
.... 3.13) $SeOF_2$	
.... 3.14) N_2O	
.... 3.15) HNO_2	

4. Answer the following questions regarding intermolecular forces.

4.1) For each of the given substances, identify their characteristics, including polar/nonpolar molecules, solubility in water, and types of intermolecular forces (1=London force, 2=Dipole-dipole force, 3=Hydrogen bond). **(12 marks)**

Substance	Polar/ Nonpolar	Soluble/ Insoluble in ammonia (NH_3)	Intermolecular force (1 or/and 2 or/and 3)
CS_2			
C_2H_5COOH			
PCl_3			
BCl_3			

4.2) Arrange the following compounds (H_2O, CS_2, C_2H_5COOH, PCl_3, BCl_3) from the highest to the lowest boiling point. To achieve full points, please explain your reasons.

(3 marks)

Boiling point: ....> ....> ....>....................>....................

Part 3 : Total score 30 marks

5. What is the difference between exothermic and endothermic process? **(6 marks)**

   (provide short answer with examples)

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6. According to the energy bond data below

$$N_{2(g)} + H_{2(g)} \rightleftharpoons NH_{3(g)}$$ (unbalanced equation)

Bond	Energy bond (kJ/mol)	Bond	Energy bond (kJ/mol)
H–H	440	N≡N	940
N–N	193	N–H	400
N=N	418		

6.1) Calculate the value of ΔH° for the reaction **(10 marks)**

6.2) Is this endothermic or exothermic process? Explain your reason **(2 marks)**

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7. Arrange the following ionic compounds in order of increasing lattice energy and explain your thoughts: NaF, CsI, and CaO. **(6 marks)**

8. Arrange the following covalent compounds in order of increasing bond length of C-C bond and provide your reasons: C_2H_2, C_2H_4, and C_2H_6. **(6 marks)**

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Dr. Nisalak Trongsiriwat

Examiner

PERIODIC TABLE OF ELEMENTS

Atomic # / Symbol / Name / Weight

C Solid · Hg Liquid · H Gas · Rf Unknown

Metals: Alkali metals · Alkaline earth metals · Lanthanoids (Lanthanides) · Actinoids (Actinides) · Transition metals · Post-transition metals
Metalloids
Nonmetals: Other nonmetals · Noble gases

Pnictogens · Chalcogens · Halogens

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H Hydrogen 1.008																	2 He Helium 4.0026
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122											5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305											13 Al Aluminium 26.982	14 Si Silicon 28.085	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)
7	87 Fr Francium (223)	88 Ra Radium (226)	89–103	104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)

For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.

6	57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97
7	89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)

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Electronegativity (EN)

1A	2A	3B	4B	5B	6B	7B	8B	8B	8B	1B	2B	3A	4A	5A	6A	7A	8A
H 2.1																	
Li 1.0	Be 1.5											B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	
Na 0.9	Mg 1.2											Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0	
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.9	Ni 1.9	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8	Kr 3.0
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5	Xe 2.6
Cs 0.7	Ba 0.9	La-Lu 1.0-1.2	Hf 1.3	Ta 1.5	W 1.7	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.9	Bi 1.9	Po 2.0	At 2.2	
Fr 0.7	Ra 0.9																